Load Bounds: Node, Edge and Lane Accounting for a Rings Node under Free Identities

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Abstract

A Rings identity is a key digest, so anyone can mint as many as they like. Every table a remote party can grow must therefore be accounted to something that is not free. We take the authenticated neighbour edge: a connection the node itself admitted, which holds one of its $\mathcal{R}$ connection slots. One accounting law covers every such table: each slot has a reserved floor, overflow shares a pool in fair claims, and the node-wide bound is the sum. An honest edge is then refused only past its slot's floor and its fair claim, and a free identity consumes at most the share of the slot that carries it. We apply the law to the receiver's replay streams (with an eviction horizon that cannot reopen replay), to the removal store, to received frames through a credit-reclaim protocol that makes the frame bound hard and independent of the connection count, and to reassembled messages, which are charged once at their first chunk so that no honest message is refused. We state the complexity obligations of the admission hot paths, and show that the barrier's control exemption is the identity "a lane waits for reassembly iff it carries chunks".

I. Problem

#907
: The receiver's replay table never evicts, and a new stream fails closed once the table is full. Streams are keyed by the origin account, which costs nothing, so 16 384 key pairs and one message each lock a node out of every new origin, permanently.

#1060, residual
: The removal store charges each sender a share, and senders are free keys, so 64 identities fill it.

#934
: Credit bounds the frames held per connection, not per node: about 1.3 GiB at the connection cap. Advertised credit cannot be taken back, so no hard node-wide bound exists today.

#932
: A reassembled message is re-charged at its full size on the inbound actor, where it cannot wait, and is dropped when that charge does not fit. A chunk that finds the reassembly budget full is dropped too.

#938
: Admission costs $O(n)$ per event and takes node-global locks per frame, exactly when the node is saturated.

#927
: It is unclear whether the barrier's control exemption is still needed under per-lane credit.

Ryan J. Kung can be reached at ryankung@ieee.org. Manuscript created October 25, 2026. This document specifies the load bounds of a Rings node before 1.0: what it holds node-wide, per neighbour edge and per lane, what a free identity can consume, and what refusal and backpressure look like at each bound (rings-transport, module callback; rings-core, modules fair_admission, swarm::callback::inbound, chunk, message::replay and dht::chord::carrier_store). It is the law for issues #907, #927, #932, #934 and #938 of RingsNetwork/rings, and closes the Sybil residual of the removal-stability law [1].

II. Model

Definition 1 (Principals).
- An origin o is an account DID, recovered from a signature. Anyone can create one, at no cost.
- An edge e is a handshake-authenticated connection the node admitted (EdgeRelation::Neighbour). The registry holds at most $\mathcal{R} = 2(F+s+1)$ connection records in any phase, where $F = 160$ is the number of finger slots and s the number of successors (328 at $s = 3$). So at most $\mathcal{R}$ edges are current at any instant. An edge identity is also a free key; what is bounded is the number of edges current at once.
- A slot $\kappa \in K = \{0, \ldots, \mathcal{R} - 1\}$ is one unit of the registry's capacity. An edge takes the least free slot $\kappa(e)$ when it is admitted and frees it when it is retired, both in the transition that announces it, so the current edges hold distinct slots. A slot is not a topology reference: one edge may back several fingers and a successor, or none.
- A lane (e, ℓ) is one of the four credit-controlled channels of an edge.
- The local principal λ is the node itself: its own operations alone (its own writes and repairs, and out-of-band connection signaling it is handed).
- A dedicated principal $d \in D_T$ is a duty of the node that remote input drives but no slot introduces, with a share of its own in table T: the inbox principal μ, which introduces deliveries out of the node's own relay inbox (replay table only), and the custody principal χ, which introduces the hand-off of a range the node owns (removal store only).

Definition 2 (Table). A table T is a node-wide set of items that remote input can grow: replay streams, removal

records, frames held, reassembly bytes, mailbox bytes. Every item $x \in T$ carries a principal $\pi(x) \in K \cup \{\lambda\} \cup D_T$: the slot of the edge that introduced it, the local principal, or a dedicated one, and it leaves T only by a transition of T's own law (consumption, retirement at a horizon, or an acknowledged move).

Assumption 1 (What a remote peer can send). A remote party can:

- mint origins without bound;
- hold up to $\mathcal{R}$ edges at once, and replace an edge by a fresh identity whenever connection admission accepts one;
- send, on each edge, anything the edge's credit, quotas and admission laws let through.

It cannot authenticate as an edge it does not hold. Sybil and eclipse resistance remain non-goals; this law bounds what a Sybil consumes, and does not prevent one.

Refusal and backpressure. At a bound, a transition takes one of three forms:

- backpressure: the transition waits for capacity in resource order, and nothing is lost;
- refusal: it fails with an error that names the bound and changes nothing, and the sender keeps its copy;
- drop: it is discarded and counted.

A drop is allowed only where the law names it (best-effort relay, the application inbox). It is never allowed for an honest frame or message at a capacity bound.

III. The accounting law

Law 1 (Slot accounting). Every table T has a floor f_T per slot, a pool $\mathcal{P}_T$, a local share L_T and a share S_d per dedicated principal $d \in D_T$, with node-wide bound

$$B_T(s) = \mathcal{R}(s)\, f_T + \mathcal{P}_T + L_T + \textstyle\sum_{d \in D_T} S_d,$$
$$\mathcal{R}(s) = 2(F + s + 1),$$

so a deployment that raises the successor count s (#774, #777) raises the floors it reserves, $2f_T$ per successor, and nothing else. Let use_κ be the number of items charged to slot κ, and $o_\kappa = \max(0, \mathrm{use}_\kappa - f_T)$ its pool items. The claimants are the slots with $o_\kappa > 0$ or a pending item over their floor; let $n \geq 1$ be their number. An item introduced by π is admitted iff:

- $\pi = \kappa$ and $\mathrm{use}_\kappa < f_T$ (the slot's floor); or
- the pool has room, $\sum o < \mathcal{P}_T$, and the claimant is within its fair claim, $o_\kappa < \lceil \mathcal{P}_T / n \rceil$; or
- $\pi = \lambda$ and $\mathrm{use}_\lambda < L_T$; or
- $\pi = d \in D_T$ and $\mathrm{use}_d < S_d$.

A transition that adds several items is admitted iff each of them is, in order, and is refused whole otherwise. A refused transition is refused or, where the law says so, it waits. An item stays charged to its slot until it leaves by T's own law, whoever holds the slot: a departure changes no charge, and an edge admitted into a slot inherits its usage. No item is ever dropped to make room: a claimant above its fair claim after n grew keeps what it holds and gains nothing until it falls below.

Theorem 1 (Isolation). Under Law 1:

1) $|T| \leq B_T$ always;
2) an item through slot κ with $\mathrm{use}_\kappa < f_T$ is never refused, however many origins, edges or slots any other party controls; in particular an edge keeps its floor for as long as it holds its slot, less what the slot held when it was admitted. The guarantee is per item: a transition adding k items through κ is admitted only if $k \leq (f_T - \mathrm{use}_\kappa) + (\lceil \mathcal{P}_T / n \rceil - o_\kappa)$ with the pool's room, and is refused whole otherwise;
3) an over-floor slot is refused only once it holds its fair claim or the pool is full, and it is guaranteed $f_T + \lceil \mathcal{P}_T / n \rceil \geq f_T + \lceil \mathcal{P}_T / \mathcal{R} \rceil$ items as the other claimants' items expire, with $n - 1$ of them;
4) the items of the slots a party holds or has held number at most $j f_T + \mathcal{P}_T$ for j such slots, and at most $\mathcal{R} f_T + \mathcal{P}_T$ however many edges it rotates.

Proof. (1) No transition moves an item between slots or into the pool: a departure changes no charge. Each slot holds at most f_T items within its floor, the pool admits only while $\sum o < \mathcal{P}_T$, the local principal holds at most L_T, and each dedicated principal at most S_d. (2) The floor test reads use_κ alone, and only transitions through κ change it. (3) A claimant below $\lceil \mathcal{P}_T / n \rceil$ is refused only with a full pool, which then holds some claimant above its claim; that claimant gains nothing and its items leave by T's law, so the pool drains to the fair claims, with $n \leq \mathcal{R}$. (4) By (1), with $j \leq \mathcal{R}$. □

Proposition 1 (Rotation residual). A party that rotates edges through free slots charges each slot it passes with up to f_T items, which stay until they leave by T's law. An honest edge later admitted into such a slot inherits that usage: it has no floor, only its fair claim, until those items leave. Charging every free slot takes at most $\mathcal{R} - c$ admissions, with c honest edges current, and an honest edge that holds its slot throughout is never affected.

Proof. Law 1: a slot's charges outlive its edge, and Theorem 1(2) for the edges that keep their slots. □

Proposition 1 is the cost this law does not remove: free keys and connection churn can leave newly admitted edges without a floor for one horizon of T per round of rotation. Only a cost on admission removes it: an admission rate limit (#1094, before 1.0) and position randomization (#780, 2.0). Charging the edge identity instead of its slot is unsound: a departing identity's floor items stay, while a fresh identity brings a fresh floor, so rotation alone grows $|T|$ past B_T.

Accounting to the origin, as the removal store did [1] and the replay table's quota does, fails part (4). An origin costs nothing, so a party controls unboundedly many, and the only bound left is the node-wide total. The slot is the unit that is both authenticated and bounded in number. The quota per origin stays: it is a rate within an edge, not a capacity.

Who overflows honestly. In an overlay of mostly browser nodes, the honest overflow is on relay edges: a browser

reached through a seed sees every origin the seed forwards to it on that one edge. Without the fair claim, a party with k edges could take the whole pool and refuse every honest relay edge for the table's horizon. With it, a relay edge keeps $\lceil \mathcal{P}_T/n \rceil$ however many slots the party holds. The alternative of charging only streams whose origin is the delivering edge, and leaving relayed streams to the origin quota, reopens #907: a relayed final-destination stream is exactly the stream a free identity creates, and the quota is per origin.

IV. Replay streams (#907)

A receiver stream s admits a transaction on two paths. Live, only while its proof is live, $t \le ts + ttl$, with $ttl \le \texttt{MAX_TTL_MS} = H$ and $ts \le t + \sigma$. Out of a relay inbox, as of its hold instant h, which the witness bounds by the sender's proof, $h \le ts + ttl$; only an application message is ever held.

Definition 3 (Relay hold horizon). A relay element held at h retires at $h + H_I$, $H_I = \texttt{MAX_RELAY_INBOX_TTL_MS}$ (7 days): every replica projects it out of its carrier at a read, a join or a sync, and a drain never delivers it. The predicate reads the signed hold instant, not the element's dot, so it is element-intrinsic and commutes with join, and a member that rewrites an old hold under a fresh dot gains nothing. An element is projected out, never refused, so a carrier with an old element stays admissible. Relay removes keep their cap.

Definition 4 (Horizon). Let $x_s = \max(ts + ttl)$ over the transactions s admitted, and $\iota(\mathsf{Application}) = H_I + \sigma$, $\iota = 0$ for every other class. The stream is spent at t iff $t \ge x_s + \iota + \sigma$. So a control, storage or E2E stream of default-lifetime traffic is spent about 10 min after its last transaction, and none later than $H + 2\sigma$; an application stream $H_I + \sigma$ later.

Lemma 1 (Eviction is sound). A spent stream can be forgotten without reopening replay. Every transaction it ever admitted is then undeliverable at this node's clock, even one stepped back by σ, on either path, so a stream opened afresh at First never sees one again.

Proof. Live, a transaction is refused at its proof once $t > ts + ttl$, and $t - \sigma \ge x_s$. Out of an inbox, it is delivered only while $t < h + H_I \le ts + ttl + H_I$, and $t - \sigma \ge x_s + H_I + \sigma$. Both checks run before replay, whatever the window holds. □

Law 2 (Stream admission). The receiver table is a table under Law 1:

- A new stream is introduced by the slot of the edge that delivered its first transaction; a delivery out of the node's own relay inbox by μ; out-of-band connection signaling by λ.
- A stream leaves the table only when it is spent and holds no relay custody, by Lemma 1. Spent streams are retired lazily, by the next admission that reads the clock past them, and their records are removed in that admission's batch.
- The record of a stream stores $(\pi(s), x_s + \iota + \sigma)$, so a restart recovers the charges.

Parameters:

- $f = 32$ new streams per slot, $\mathcal{P} = 4096$, $L = 1024$ and $M = S_\mu = 768$, so $B(3) = 328 \cdot 32 + 4096 + 1024 + 768 = 16\,384$, the table's hard capacity, and $B(s) = B(3) + 64(s-3)$: past $s = 3$ the hard capacity binds first;
- the per-origin quota is unchanged.

A new stream past its bound is refused with TransactionReplayStreamCapacityExceeded, as today. The refusal now lasts until a stream of the refused slot is spent, at most $H + 2\sigma$ (100 min, about 10 min for default-lifetime traffic) and $H + H_I + 3\sigma$ for application streams, and it reaches only a slot past its floor and its fair claim.

Proposition 2 (No permanent lockout). A party with k slots and any number of origins can refuse new streams only through those k slots, and only once the pool is full. For every other slot, a new origin is refused once that slot holds $f + \lceil \mathcal{P}/n \rceil$ unspent streams. Every refusal ends within $H + 2\sigma$, or $H + H_I + 3\sigma$ for an application stream, of the last admission on the streams that fill the bound.

Proof. Theorem 1, and Lemma 1 for the end of a refusal. □

The application horizon is the price of offline mail: a stream must outlive every hold of its messages. It lengthens what Proposition 1 costs an honest newcomer in an application-charged slot to H_I, not what any principal can hold.

Inbox mail draws on μ, never on L, so a flood of mail cannot refuse this node's own signaling. The residual: once mail from M distinct origins is unspent, mail from a new origin is refused at delivery for up to $H_I + H + 3\sigma$; its element stays held, unless its own hold horizon passes first. A slot would not do better: a node's inbox mail all enters through its own store or one member's fetch reply, so charging the delivering slot isolates nothing.

A relay or seed that forwards many distinct origins can fill its own floor legitimately; its overflow then holds a fair claim of the pool, $\lceil \mathcal{P}/n \rceil \ge 13$ at worst and 4096 alone, and past it the honest cost is a refusal on that edge until its streams are spent. The sender table holds this node's own streams, so it is local.

V. The removal store, re-keyed

Law 6 of [1] charged each remove to its sender's origin. We re-key it under Law 1:

- an operation or a sync is charged to the slot of the edge that delivered it, and this node's own writes and repairs to λ; a read repair of a remote lookup reply is charged to the slot that delivered the reply, so lookup replies never draw on L;
- a hand-off sync of a placement this node's view makes it own, from the node its view names as the placement's

previous owner (its predecessor or a successor), is charged to the custody principal χ; every other sync from a predecessor or successor is charged to its slot, as any edge's.

Parameters: $f = 128$ removes per slot, $\mathcal{P} = 16\,384$, $L = 7\,168$ and $K = S_\chi = 16\,384$, so $B(3) = 328 \cdot 128 + 16\,384 + 7\,168 + 16\,384 = 81\,920$, about 12 MB, and $B(s) = B(3) + 256(s-3)$. Theorem 1 then closes the Sybil residual that #907 was to close: one mechanism serves both tables.

A store admits one carrier per transition, at most $T_{\max} = 1024$ removes, so all-or-none binds per topic. The binding limit of a hand-off is cumulative: without χ, every topic of a range a neighbour hands over draws on that neighbour's one slot, $f + \lceil \mathcal{P}/n \rceil$ in all, and a busy range (about 10 removes a minute per topic stays ≈ 1000 unstable) would wait up to $H + \sigma$ while the new owner misses its reads. The residuals of χ: a Byzantine predecessor or successor can fill K (a non-goal, as the holder class was), and an honest range of more than K unstable removes still waits for them to stabilize.

VI. Received frames (#934)

Let lane (e, ℓ) have advertised credit A and released count R. Its liability is $\text{liab} = A - R$: the frames the receiver may still have to hold, in flight or held. Today $\text{liab} \le W = 16$ per lane, but credit is cumulative and never shrinks, so $\sum \text{liab}$ is bounded only by the connection count.

Definition 5 (Reclaim). Two frames are added on the priority lane, both cumulative and idempotent:

- $\mathsf{Shrink}(\ell, L', g)$, receiver to sender: "from generation g, your limit on lane ℓ is L'".
- $\mathsf{Returned}(\ell, c, g)$, sender to receiver: "I committed c frames of ℓ, and from now on I commit at most $\max(c, L')$".

Until the Returned arrives, the receiver counts the lane's liability at its old A. After it, the liability is $\max(c, L') - R$.

Law 3 (Hard frame bound). Every lane has a floor g_ℓ: $g_{\text{prio}} = 2$ for the priority lane (control, and the reclaim and demand frames), and $g_\ell = 1$ for each of the three data lanes, so the first frame of a burst on any lane never waits a round trip. A fresh generation grants only the floors.

- A data lane with queued frames and no credit sends a one-frame $\mathsf{Wanted}(\ell)$ on the priority lane; a receiver that admits a lane's floor frame also grants it pooled credit at once, so a burst's second frame waits at most the round trip its first frame already took.
- The receiver grants a lane credit beyond its floor only from the node's frame pool:

$$\sum \text{liab} \le B_F(s) = \mathcal{R}(s)\,(g_{\text{prio}} + 3g_{\text{data}}) + \mathcal{P}_F.$$

- Demands are served first come, first served.
- A lane that received no frame for T_{idle} while holding pooled credit is shrunk to its floor.

Parameters:

- $\mathcal{P}_F = 256$ frames (16 MiB, the soft limit of today);
- T_{idle} = two observed round trips;
- $B_F(3) = 328 \cdot 5 + 256 = 1\,896$ frames, 118.5 MiB, and $B_F(s) = B_F(3) + 10(s-3)$ frames.

This is independent of how many connections are open. Data-lane floors of 0 would lower it to 57 MiB, at the cost of one round trip before the first frame on any idle lane: on an onion loop (#843) that is up to one round trip per hop whose lane went idle, on circuit setup and on the first cell after a pause. We keep the floor at 1, and the onion data plane pays nothing.

Proposition 3 (Frames). Under Law 3, for an honest sender:

1) $\sum \text{liab} \le B_F$ in every interleaving, with shrink, return and demand frames reordered and duplicated;
2) no honest frame is refused: a frame sent under credit is within the liability counted for it;
3) a lane with queued frames is never blocked forever while the receiver keeps consuming (progress);
4) an idle lane holds at most its floor after T_{idle} plus one round trip, so it starves no active lane, and its first frame after the pause is sent at once;
5) control traffic never waits for the pool.

Proof sketch. (1) Pooled credit is granted only when the counted liability fits, and the counted liability never falls below the true one: a shrink takes effect only once its Returned confirms what was committed. (2) A frame committed under A is counted until released. (3) A blocked lane has a Wanted queued. The pool refills as frames are released and idle lanes are shrunk, and service is first come, first served. (4) The shrink frees the lane's pooled credit. (5) The priority floor is reserved per edge. These are the model-checking obligations of the implementation, over the existing credit model with the new frames. □

This is a wire change: a total cutover, with no versioning before 1.0. Byte-denominated credit (#937) composes with it, by counting liab in bytes, and is not required here.

VII. Reassembled messages (#932)

Law 4 (One charge per message). The first chunk of a message, with header $[0, n]$ and data length d_0, reserves $m = n \cdot d_0$ (validated $\le$ max_message_bytes) once, on the inbound mailbox ledger: the edge's budget, the lane's share and the shared pool. It takes that reservation as an arrival, so it waits in resource order and is never refused for capacity.

- Later chunks of the message must have length $\le d_0$, and they charge nothing.
- The reassembly buffer and the completed message live inside that one reservation.
- The mailbox transition at completion is a move, not a new charge.
- The reservation is released when the message's processing completes, or when the reassembly expires.

The separate reassembly byte budget and its refusal (Rejected(Capacity)) are retired. The id and slot caps stay, as validity bounds.

Proposition 4 (No honest message refused). Under Law 4, no honest chunked message is refused or dropped for capacity, and no wait can deadlock.

Proof. The only charge is the first chunk's, which waits as an arrival before it holds a lane ticket. The ordering laws of ResourceOrderedQueue cover it, and the capacity it waits for is released by events that do not wait on it. The transition that used to re-charge, and could only be refused, now moves bytes already reserved. □

The cost is direction 1 of #932: a message holds its full size for its whole reassembly. This is bounded by the edge's budget and by the reassembly expiry, so a peer that announces and then stalls holds only its own budget.

VIII. Hot paths (#938)

These are performance obligations, kept apart from the security laws above.

Obligation 1 (Admission complexity).
- Each arrival, release and cancel in ResourceOrderedQueue costs $O(\log n)$ in the waiters: waiters are indexed by resource, and one ledger snapshot is taken per pass.
- Each received frame takes $O(1)$ lock acquisitions, and none of them on a node-global mutex: the node load is an atomic counter.
- The node load has hysteresis: advertisements are deferred above H_{hi} and resume below $H_{\mathrm{lo}} = H_{\mathrm{hi}}/2$. Deferred links are kept in a set, so a release at the limit wakes no one, and crossing H_{lo} wakes each deferred link once.
- Each peer runs one admission watcher, not one task per send.
- Unread admissions are indexed by peer.

They are verified against the sync-storm matrix, before and after each change.

IX. The barrier exemption (#927)

Lemma 2 (One frame per lane). If every path that hands a frame to core does so through its lane's FIFO, with at most one frame in hand-over per lane, then a lane that waits behind the reassembly barrier waits at most one reassembly frame's service, $\leq 2\,\mathrm{s}$. That is below the control deadline of 15 s.

The predicate lane_waits_for_reassembly is "the lane carries logical data". Chunks travel only on their message's lane (the lane-mismatch refusal), and control messages are never chunked, so the predicate is exactly "the lane can carry chunks". In that form it is the barrier's definition, not an exemption.

Decision. Keep the predicate, renamed for what it is. Remove the ablation toggle ProtectionLayer::BarrierControlExemption and its legacy feedback-loop witness. This is a subtraction, under two obligations checked in the implementation:

- every hand-off path obeys Lemma 2, the pre-admission drain and the detached submissions included;
- no control message can exceed one frame. This is asserted at compile time from the widest control message.

X. Summary of bounds

Table	Bound	At the bound
frames held	node $B_F(s)$; lane floor	sender waits for credit (backpressure)
mailbox bytes	edge budget, lane share, pool	arrival waits (backpressure)
reassembly	inside the mailbox reservation	first chunk waits (backpressure)
replay streams	$\mathcal{R}(s)f + \mathcal{P} + L + M$ by slot, fair pool	new stream refused, $\leq H + 2\sigma$ ($H + H_I + 3\sigma$ application)
removal store	$\mathcal{R}(s)f + \mathcal{P} + L + K$ by slot, fair pool	write refused, unacked
origin	rate quota within its edge	refused for the rate window
application inbox	64 MiB	dropped, counted (by design)

What a free identity consumes. An origin alone consumes nothing beyond its quota on the edge that carries it. An edge consumes its slot's floors, which outlive it until its items leave. The pool is shared, and is the most that any number of free identities can take together.

XI. Implementation order

PR	Issue	Law
1	#907	Laws 1, 2; Sec. V
2	#932	Law 4
3	#934	Law 3 (wire cutover)
4	#938	Obligation 1, measured
5	#927	Sec. IX (subtraction)

PR 1 adds the slot-accounting primitive that PR 2 reuses for the mailbox, and the relay element horizon of Definition 3. PR 4 rewrites the queue that PR 2 changes, so it follows PR 2. PR 5 needs nothing from the others.

Operator steps, in the CHANGELOG under Unreleased:

- PR 1 changes the replay record, which gains its introducer and spent instant. Remove the transaction-replay store while the node is stopped; this is the documented reset of the durable-records law. The removal store gains a slot charge, so remove dht-removals too. Held mail now expires H_I after its hold, even in a carrier that newer holds keep alive.
- PR 3 is a wire cutover with no step.

References

[1] R. J. Kung, "Removal stability: Eviction, tombstone bounds and heartbeats for the data topics of Rings," papers/removal-stability.tex, https://github.com/RingsNetwork/rings, 2026, issue #1060.